Medicine, Dentistry and Medical laboratory

Cell Biology

Lecture 2

Biology of Living Cell

What is the meaning of cell biology?

The cell (from Latin cella, meaning small room) is the basic structural, functional and biological unit of all known living organisms.

* The study of cells is called Cell Biology (Cytology).

Types of microscopes:

1- Light microscope.
2- Electron microscope.

The sizes of living things and their components:

The unassisted human eye can usually see macroscopic organisms and a few large cells.

Microscopic cells are visible with the light microscope, but not in much detail. An electron microscope is necessary to see organelles in detail and to observe viruses and molecules.

The invention of the microscope, which allowed scientist for the first time to see biological cell.
Antoni van Leeuwenhoek (1632–1723) may have been the first person to see living cells by the use of the microscope.
In this way, he reported seeing the parasite ***Giardia lamblia*** in his feces.

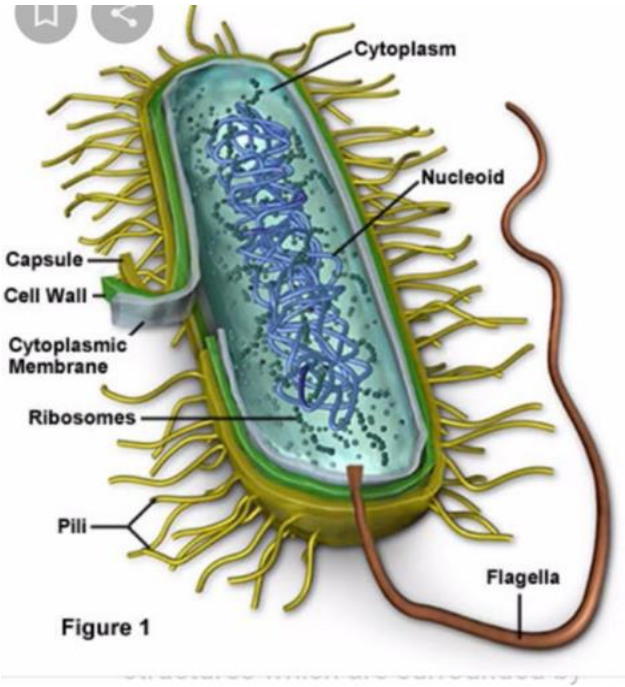
Giardia lamblia

Cell theory:
The cell theory is based upon the work of Schleiden, Schwann, Virchow and many other scientist.
It states that:

1- All organisms are composed of cells.
2- Cells are the basic units of structure and function in organisms.
3- Cells come only from preexisting cells because cells are self-reproducing.

There are two different types of cells exist:
1- Prokaryotic cells:
Which lack a membrane-bounded nucleus and the DNA (deoxyribonucleic acid) is located in a region called a nucleoid (e.g. Bacteria).

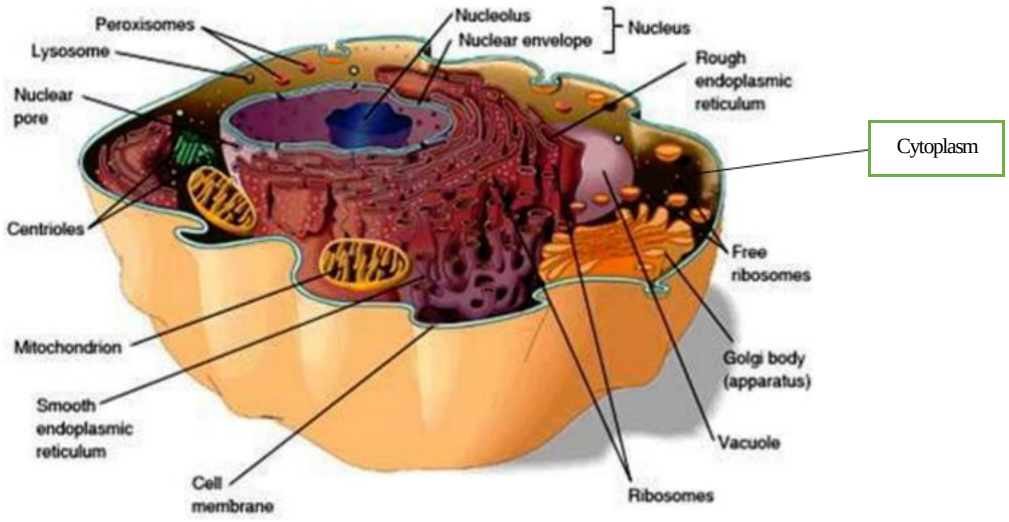
* They are present in great numbers in the air, in bodies of water, in the soil, and they also live in and on other organisms. Although they are structurally less complicated.



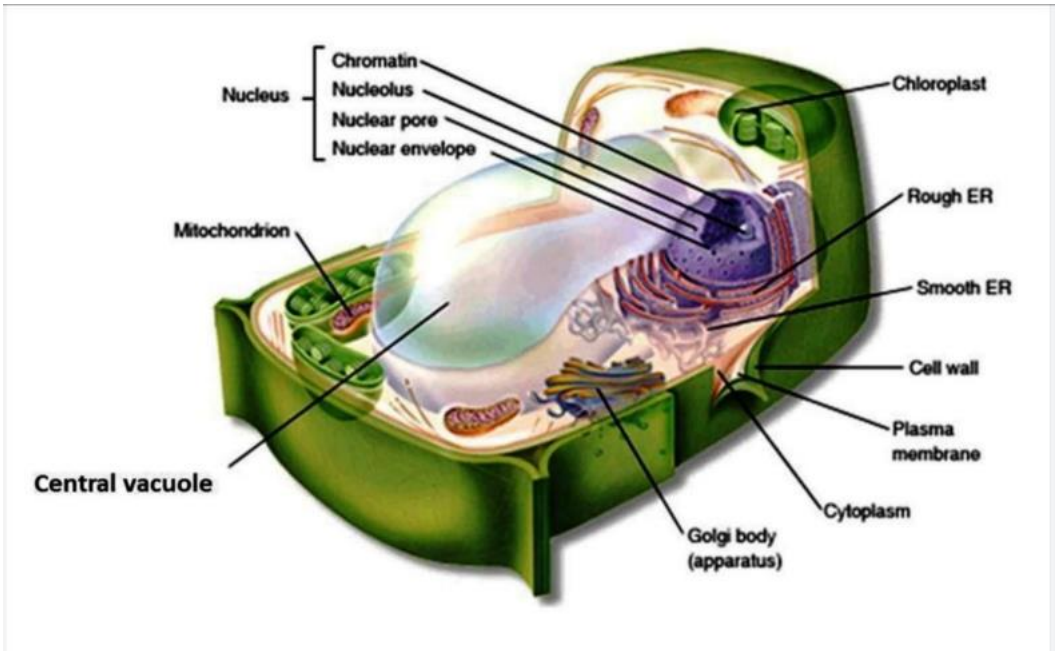
Bacterial cell

2- Eukaryotic cells:

They are defined as cells containing organized nucleus and organelles which are enveloped by membrane-bound organelles.
Examples of eukaryotic cells: animal cells and plant cells.



Animal cell



Plant cell

* All living organisms are composed of cells, they may have modifications to suit their way of life (adaptation).

Classification of living organisms:

In new classification, they are subdivided into **three** major **Domains:** Archaea, Bacteria and Eukarya.

Each domain is further subdivided into separate Kingdoms.

Domains of life:

1- Domain : Archaea

*Kingdom : Archaeabacteria

2- Domain : Bacteria

*Kingdom : Eubacteria

3- Domain : Eukarya

* Kingdom : Protista

*Kingdom : Fungi

*Kingdom : Plantae

*Kingdom : Animalia